IMPLEMENTAREA CONCURENTEI IN LIMBAJE DE PROGRAMARE

Concurenta

Threaduri

Memorie Partajata

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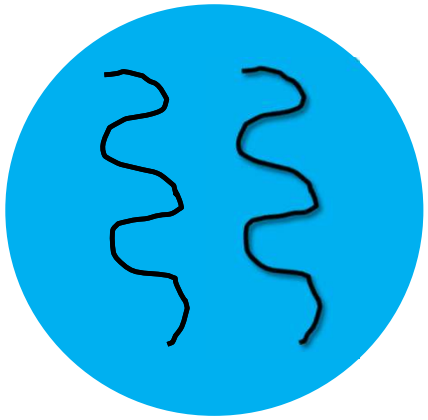
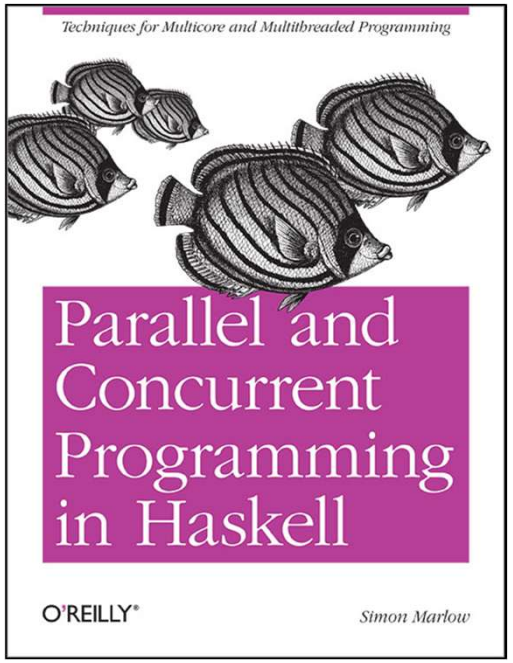


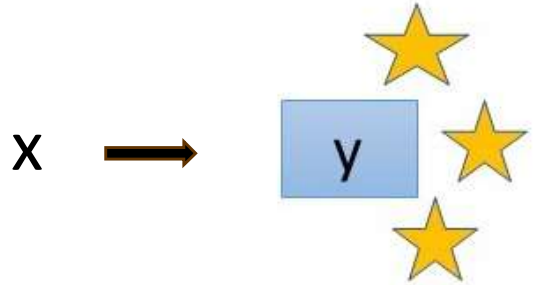


Part II. Concurrent Haskell
Cap.7 & 8

https://hackage.haskell.org/package/base-4.20.0.1/docs/Control-Concurrent.html

➢Monade in Hakell

Intuitie:
functii care calculeaza o valoare si produc un efect



Monada este o clasa de tipuri

```
class Applicative m => Monad m where
   (>>=) :: m a -> (a -> m b) -> m b
   (>>) :: m a - > m b -> m b
   return :: a -> m a

   ma >> mb = ma >>= \_ -> mb
```

- **m a** este tipul computatiilor care produc rezultate de tip **a** si au efecte laterale
- **a –> m b** este tipul continuarilor / al functiilor cu efecte laterale
- **>>=** este operatia de „secventiere” a computatiilor
- **return** este o functie care produce efectul nul

do notation

```
x  <- e                e >>= \x -> rest
 rest

e                      e >>= \_ -> rest
rest

e                      e >> rest
rest
```

De exemplu

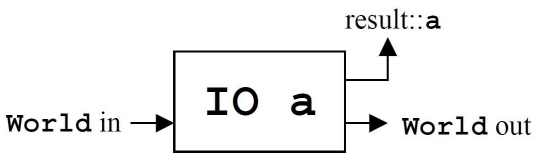
```
e1   >>= \x1 ->
e2   >> e3
```

devine

```
do
    x1 <- e1
    e2
    e3
```

➢ Monada IO

Intrarile si iesirile sunt valori de tipul IO a




```
Prelude> :t getLine
getLine :: IO String
Prelude> :t putStrLn
putStrLn :: String -> IO ()
```

```
Prelude> :t getChar
getChar :: IO Char
Prelude> :t putChar
putChar :: Char -> IO ()
```

```
Prelude> :t print
print :: Show a => a -> IO ()
```

() unit este singura valoare a tipului () (singleton)

IO () este folosit atunci cand actiunile nu intorc valori semnificative

Valoarea de tip a dintr-o valoare de tip IO a se obtine folosind <-

```
s <- getLine
c <- getChar
```

➢ Actiuni IO

O valoare de tip (IO a) este o actiune care, daca este executata, produce o data de tip a.

Actiunile se comporta asemenea instructiunilor.
Secventele de actiuni de obtin folosind notatia do.

```
pg = do  putStrLn "Numele"
         s <- getLine
         putStrLn ("Hello " ++ s)
```

```
*Main> pg
Numele
Ioana
Hello Ioana
*Main> :t pg
pg :: IO ()
```

➢ Blocul do

In general un bloc do poate fi descris ca o secventa de Instructiuni, iar o instructiune poate fi:

- actiune, adica o expresie de tipul IO (de ex: getLine)
- o legatura <- (de ex: s <- getLine)
- o declaratie let (fara in)
- un apel al functiei return

```
pg = do
        putStrLn "introdu sirul"
        s <- getLine
        let n = length s
        putStrLn (s ++ " are " ++ (show n) ++ " litere")
```

```
*Main> pg
introdu sirul
abcd
abcd are 4 litere
```

➢ **Functii monadice:** **sequence, sequence_, mapM, mapM_, foldM, foldM_**

```
Prelude Control.Monad> [l1,l2,l3] <- sequence [getLine, getLine, getLine]
linia 1
linia 2
linia 3
Prelude Control.Monad> l2
"linia 2"
```

```
Prelude Control.Monad> mapM print [1,2,3]
1
2
3
[(),(),()]
Prelude Control.Monad> mapM_ print [1,2,3]
1
2
3
```

```
Prelude Control.Monad> let g = \x y -> (putStrLn  (x ++ show y)) >> (return $ (x ++ show y))
Prelude Control.Monad> let z = g "a" 4
Prelude Control.Monad> z
a4
"a4"
Prelude Control.Monad> foldM g "a" [1,2,3]
a1
a12
a123
"a123"
```

- Variabile mutabile: IORef

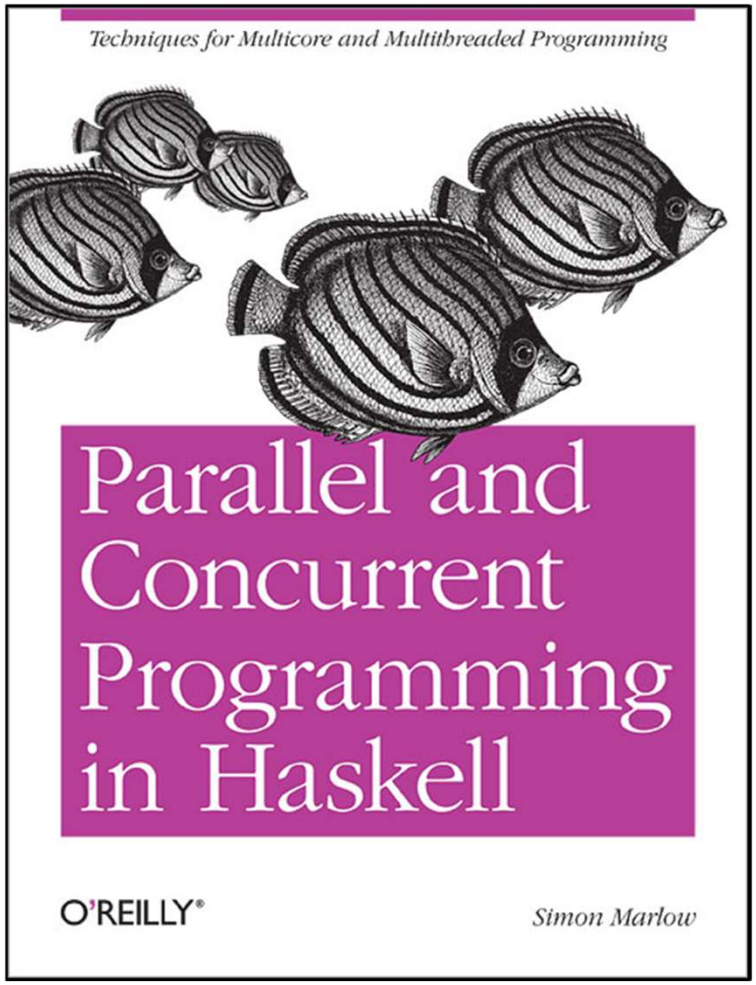
```
import Data.IORef
-- variabile mutabile in monada IO

newIORef :: a -> IO (IORef a)
readIORef :: IORef a -> IO a
writeIORef :: IORef a -> a -> IO ()
```

```
add :: IORef Int -> Int ->  IO()
add rref  n = do
            val <- readIORef rref
            writeIORef rref

main = do
        rref <- newIORef 0
        add rref 10
        val <- readIORef rref
        print val
```

"Haskell does not take a stance on which concurrent programming model is best: actors, shared memory, and transactions are all supported, for example."

"Haskell provides all of these concurrent programming models and more - but this flexibility is a double-edged sword. The advantage is that you can choose from a wide range of tools and pick the one best suited to the task at hand, but the disadvantage is that it can be hard to decide which tool is best for the job."

"Therefore, to learn Concurrent Haskell, we can start from the low-level interfaces and then explore how to combine them and build on top to create higher-level abstractions, which is exactly the approach taken in this book. The aim is that by building up the implementations of higher-level abstractions using the low-level features, the higher-level abstractions will be more accessible and less mysterious than if we had just described an API. Furthermore, by seeing examples of how to build higher-level abstractions, you should be able to go away and build your own variations or entirely new libraries."



Thread-urile in Haskell:

- Thread-urile sunt "leighweight"

" Concurrency is "lightweight", which means that both thread creation and context switching overheads are extremely low. Scheduling of Haskell threads is done internally in the Haskell runtime system, and doesn't make use of any operating system-supplied thread packages."

https://hackage.haskell.org/package/base-4.21.0.0/docs/Control-Concurrent.html

- Implementarea ofera garantii de corectitudine

"We would like our concurrent programs to be executed with some degree of *fairness*. At the very least, no thread should be starved of CPU time indefinitely, and ideally each thread should be given an equal share of the CPU. GHC uses a simple round-robin scheduler. It does guarantee that no thread is starved indefinitely, although it does not ensure that every thread gets an exactly equal share of the CPU. In practice, though, the scheduler is reasonably fair in this respect."

https://smunix.github.io/chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_fairness

➢ Programarea concurenta in Haskell are loc in monada IO.

forkIO :: IO () -> IO ThreadId

crearea thread-urilor este o actiune IO

```
Prelude> :m + Control.Concurrent
Prelude Control.Concurrent> :t forkIO
forkIO :: IO () -> IO ThreadId
```

```
import Control.Concurrent
import Control.Monad

main = do
        forkIO (replicateM_  100 (putChar 'A'))   -- child thread
        replicateM_ 100 (putChar 'B')  -- main thread
```

thread-ul principal este "bounded", adica corespunde unui thread al sistemului de operare.

replicateM

```
Prelude> :m + Control.Monad
Prelude Control.Monad> :t replicateM_
replicateM_ :: Monad m => Int -> m a -> m ()
Prelude Control.Monad> replicateM_ 5 (putStrLn "A")
A
A
A
A
A
```

forkIO :: IO () -> IO ThreadId

```
import Control.Concurrent
import Control.Monad

main = do
        forkIO (replicateM_ 100 (putChar 'A'))   -- child thread
        replicateM_ 100 (putChar 'B')  -- main thread
```

```
*Main Control.Monad> main
ABABABABABABABABABABABABABABABABABABABABABABABAB
BABABABABABABABABABABABABABABAB
```

La rulari diferite se pot obtine rezultate diferite!

forkIO :: IO () -> IO ThreadId

```
import Control.Concurrent
import Control.Monad

main = do
        forkIO (replicateM_  100 (putChar 'A'))   -- child thread
        replicateM_ 100 (putChar 'B')  -- main thread
```

Daca fisierul se numeste *thread.hs* atunci el poate fi compilat si executat:

ghc thread.hs -threaded
thread

```
C:\Users\igleu\Documents\DIR\ICLP\ICLP2023\haskell2023>ghc thread.hs -threaded
Loaded package environment from C:\Users\igleu\AppData\Roaming\ghc\x86_64-mingw32-8.10
.7\environments\default

C:\Users\igleu\Documents\DIR\ICLP\ICLP2023\haskell2023>thread
BBABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABAB
ABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABAB
ABABABABABABABABABABABABAA

C:\Users\igleu\Documents\DIR\ICLP\ICLP2023\haskell2023>thread
BBABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABAB
ABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABABAB
ABABABABABABABABABABABABABAA
```

La rulari diferite se pot obtine rezultate diferite!

forkIO :: IO () -> IO ThreadId

myThreadId :: IO ThreadId

```
import Control.Concurrent
import Control.Monad

main = do
        forkIO (replicateM_  100 (myThreadID >>= print))   -- child thread
        replicateM_ 100 (myThreadID >>= print)  -- main thread
```

```
PS C:\Users\igleu\Documents\DIR\ICLP22\Curs-2022\Haskell22\pgh\haskell2022> ./threadID
ThreadId 3
ThreadId 4
ThreadId 3
ThreadId 4
ThreadId 3
ThreadId 4
ThreadId 3
ThreadId 4
ThreadId 3
ThreadId 4
ThreadId 3
ThreadId 4
```

"The computation passed to forkIO is executed in a new thread that runs concurrently with the other threads in the system. If the thread has effects, those effects will be interleaved in an indeterminate fashion with the effects from other threads."
S. Marlow, PCPH

"forkIO is assymetrical: when a process executes a forkIO it spawns a child process that executes concurrently with the continued execution of the parent"
SL Peyton Jones, A Gordon, S Finne, Concurrent Haskell

"**GHC's runtime system treats the program's original thread of control differently from other threads. When this thread finishes executing, the runtime system considers the program as a whole to have completed. If any other threads are executing at the time, they are terminated.**"
B. O'Sullivan, D. Stewart, J. Goerzen, Real World Haskell

- Interleaving

```
import Control.Concurrent
import Control.Monad

myread1 = do
        putStrLn "thread1"
        s<- getLine
        putStrLn $ "cititt 1: " ++ s

myread2 = do
        putStrLn "thread2"
        s<- getLine
        putStrLn $ "cititt 2:" ++ s


main = do
          forkIO (replicateM_  10 myread1)
          replicateM_ 10 myread2
```

```
*Main> main
thread1
thread2
e
cititt 1: e
thread1
s
cititt 2:s
thread2
r
cititt 1: r
thread1
e
cititt 2:e
thread2
f
cititt 1: f
thread1
f
```

- Executie secventiala vs executie concurenta

```
fib 0 =1
fib 1 =2
fib n = fib (n-1) + fib (n-2)

act n =  do
        let x = (fib n)
        putStrLn ("Fib " ++ (show n) ++ " is " ++ (show x))

act4 = do
      act 10
      act 20
      act 30
      act 35
      getLine
```

```
main = do
      forkIO $ act 10
      forkIO $ act 20
      forkIO $ act 30
      forkIO $ act 35
      getLine
```

- Executie secventiala vs executie concurenta

```
act4 = do
    act 10
    act 20
    act 30
    act 35
    getLine

main = do
    forkIO $ act 10
    forkIO $ act 20
    forkIO $ act 30
    forkIO $ act 35
    getLine
```

```
Prelude> :set +s
Prelude> :l test.hs
[1 of 1] Compiling Main             ( test.hs, interpreted )
Ok, one module loaded.
(0.09 secs,)
*Main> act4
Fib 10 is 144
Fib 20 is 17711
Fib 30 is 2178309
Fib 35 is 24157817

""
(115.38 secs, 11,248,521,824 bytes)
*Main> main
FiFFFbiii bbb1   0233 005i   siii sss1   4147
711
2178309
24157817

""
(114.36 secs, 57,352 bytes)
```

Atentie!
Accesul la stdout nu este thread-safe, deci trebuie sincronizat

- Compilare cu -threaded

```
act4 = do
    act 10
    act 20
    act 30
    act 35
    getLine

main = do
    forkIO $ act 10
    forkIO $ act 20
    forkIO $ act 30
    forkIO $ act 35
    getLine
```

```
*Main> :! ghc --make -threaded test.hs
```

programul este compilat cu optiunea **-threaded**

```
> .\test +RTS -N4 -s
Fib 10 is 144
Fib 20 is 17711
Fib 30 is 2178309
Fib 35 is 24157817


  INIT    time    0.000s  (  0.001s elapsed)
  MUT     time    4.672s  (  7.394s elapsed)
  GC      time    0.156s  (  0.120s elapsed)
  EXIT    time    0.000s  (  0.001s elapsed)
  Total   time    4.828s  (  7.515s elapsed)
```

-threaded

"Link the program with the “threaded” version of the runtime system.

The threaded runtime system […] enables the -N ⟨x⟩ RTS option to be used, which allows threads to run in parallel on a multiprocessor or multicore machine.

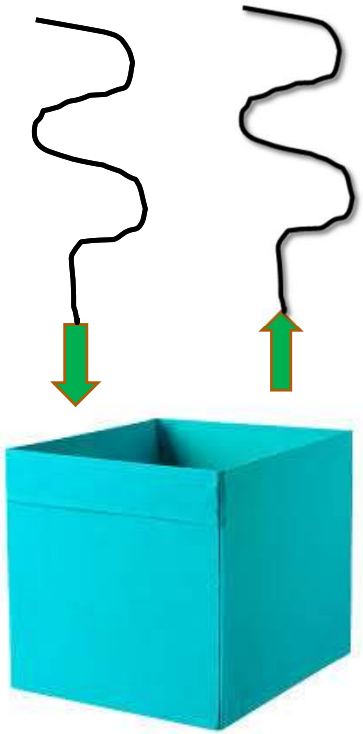
Note that you do not need -threaded in order to use concurrency; the single-threaded runtime supports concurrency between Haskell threads just fine."

https://downloads.haskell.org/~ghc/9.10.1/docs/users_guide/phases.html

"**Concurrent Haskell** is a collective name for the facilities that Haskell provides for programming with multiple threads of control. Unlike parallel programming, where the goal is to make a program run faster by using more CPUs, the goal in concurrent programming is usually to write a program with multiple interactions."

S. Marlow, PCHP

- Comunicarea thread-urilor



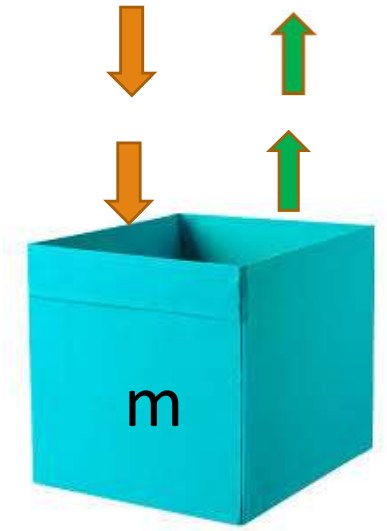
MVar

mutable variable

➢ Comunicarea folosind MVar se face in monada IO

data MVar a

- o data de tipul MVar a reprezinta o locatie mutabila care poate fi goala sau
- poate contine o singura valoare de tip a
- thread-urile pot comunica prin intermediul datelor de tip MVar



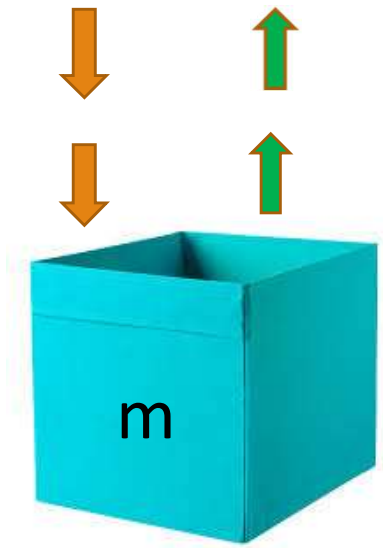
m :: MVar a

poate fi vazuta ca:

- un semafor binar
- un monitor cu o variabila

➢ Comunicarea folosind MVar se face in monada IO

```
data MVar a

newEmptyMVar :: IO (MVar a)  -- m <- newEmptyMVar
                             -- m este o locatie goala

newMVar :: a -> IO (MVar a)   -- m <- newMVar v
                              -- m este o locatie care contine valoarea v

takeMVar :: MVar a  -> IO a   -- v <- takeMVar m
                              -- intoarce in v valoarea din m si goleste m
                              -- asteapta (blocheaza thread-ul)  daca m este goala

putMVar :: Mvar a -> a -> IO()  -- putMVar m v
                                -- pune in m valoarea v
                                -- asteapta (blocheaza thread-ul) daca m este plina
```



➢ takeMVar

- takeMVar este o operatie care blocheaza thread-urile

- takeMVar este *single-wakeup*:
  daca variabila MVar este.. goala, toa.te thread-urile care vor sa execute takeMVar sunt blocate; cand variabila devine plina, un singur thread este trezit si acesta va executa takeMVar

- daca mai multe thread-uri sunt blocate pe acelasi MVar, ele vor fi trezite in ordinea FIFO

https://www.haskell.org/hoogle/?hoogle=MVar

➢ takeMVar vs readMVar

readMVar

Citeste atomic continutul unui MVar.

Daca variabile MVar este goala, thread-ul care apeleaza readMVar va astepta pana cand MVar primeste o valoare si va citi valoarea pusa de urmatoarea operatie putMVar.

readMVar este *multiple-wakeup*, deci toate thread-urile care asteapta sa citeasca din MVar vor fi trezite in acelasi timp.

Implementarea veche (intuitiva)

```
readMVar :: MVar a -> IO a
readMVar m =   do
                a <- takeMVar m
                putMVar m a
                return a
```

Implementarea actuala garanteaza ca readMVar este o operatie atomica.

https://www.haskell.org/hoogle/?hoogle=MVar

- Variabile mutabile: IORef, MVar

```
import Data.IORef
-- variabile mutabile in monada IO

newIORef :: a -> IO (IORef a)
readIORef :: IORef a -> IO a
writeIORef :: IORef a -> a -> IO ()
```

```
add :: IORef Int -> Int ->  IO()
add rref  n = do
            val <- readIORef rref
            writeIORef rref

main = do
        rref <- newIORef 0
        add rref 10
        val <- readIORef rref
        print val
```

```
import Control.Concurrent.MVar
-- variabile de sincronizare
-- variabile mutabile in monada IO

newEmptyMVar :: IO (MVar a)
newMVar :: a -> IO (MVar a)
takeMVar :: MVar a -> IO a       -- blocheaza thread-ul
putMVar :: MVar a -> a -> IO () -- blocheaza thread-ul
```

```
import Control.Concurrent

main = do
        m <- newEmptyMVar
        forkIO $ do
                putMVar m 'x'
                putMVar m 'y'
        x <- takeMVar m
        print x
        x <- takeMVar m
        print x
```

```
*Main> main
'x'
'y'
```

newEmptyMVar :: IO (MVar a)

putMVar :: MVar a -> a -> IO()

takeMVar :: MVar a -> IO a

```haskell
import Control.Concurrent

main = do
        m <- newEmptyMVar
        takeMVar m
```

```
*Main> main
*** Exception: thread blocked indefinitely in an MVar operation
```

- Sincronizare : doua thread-uri incrementeaza acelasi contor
  vrem sa citim valoarea contorului dupa ce ambele thread-uri au terminat

```
add m  =  replicateM_ 1000 $ do
                                x <- takeMVar m
                                putMVar m (x +1)

main = do
        m <- newMVar 0
        forkIO (add m )
        forkIO (add m )
        x  <- takeMVar m
        print x
```

- Sincronizare : doua thread-uri incrementeaza acelasi contor
  vrem sa citim valoarea contorului dupa ce ambele thread-uri au terminat

```
add m  =  replicateM_ 1000 $ do
                                x <- takeMVar m
                                putMVar m (x +1)

main = do
        m <- newMVar 0
        forkIO (add m )
        forkIO (add m )
        x  <- takeMVar m
        print x
```

```
*Main> :l cont.hs
[1 of 1] Compiling Main
Ok, one module loaded.
*Main> main
0
*Main> main
0
```

- Sincronizare : doua thread-uri incrementeaza acelasi contor
  vrem sa citim valoarea contorului dupa ce ambele thread-uri au terminat

```
add m  =  replicateM_ 1000 $ do
                                x <- takeMVar m
                                putMVar m (x +1)

main = do
        m <- newMVar 0
        forkIO (add m )
        forkIO (add m )
        x  <- takeMVar m
        print x
```

```
*Main> :l cont.hs
[1 of 1] Compiling Main
Ok, one module loaded.
*Main> main
0
*Main> main
0
```

trebuie sa ne asiguram ca ambele thred-uri au terminat

- Sincronizare

```
add m  ms1 = do
                replicateM_ 1000 $ do
                        x <- takeMVar m
                        putMVar mv (x +1)
                putMVar ms1 "ok"
```

```
main = do
        m <- newMVar 0
        ms1 <- newEmptyMVar
        ms2 <- newEmptyMVar
        forkIO (add m ms1)
        forkIO (add m ms2)
        takeMVar ms1
        takeMVar ms2
        x  <- takeMVar m
        print x
```

variabilele ms1 si ms2 actioneaza ca niste semafoare ; astfel ne asiguram ca ambele thread-uri au terminat

- Sincronizare

```
add m  ms1 = do
                replicateM_ 1000 $ do
                        x <- takeMVar m
                        putMVar mv (x +1)
                putMVar ms1 "ok"
```

```
main = do
        m <- newMVar 0
        ms1 <- newEmptyMVar
        ms2 <- newEmptyMVar
        forkIO (add m ms1)
        forkIO (add m ms2)
        takeMVar ms1
        takeMVar ms2
        x  <- takeMVar m
        print x
```

```
*Main> main
2000
```

- Sincronizare: doua thread-uri incrementeaza acelasi contor

threadDelay nr
suspenda thread-ul pt.nr microsecunde

```
main = do
        m <- newMVar 0
        ms1 <- newEmptyMVar
        ms2 <- newEmptyMVar
        forkIO (add1 m ms1)
        forkIO (add2 m ms2)
        takeMVar ms1
        takeMVar ms2
        x  <- takeMVar m
        print x
```

```
add1 m  ms1 = do
        replicateM_ 1000 $ do
                x <- takeMVar m
                threadDelay 100 –nu afecteaza sincronizarea
                putMVar mv (x +1)
        putMVar ms1 "ok"
```

Sincronizarea este asigurata de faptul ca ambele thread-uri apeleaza intai takeMVar

```
add2 m ms2   = do
        replicateM_ 1000 $ do
                s<- takeMVar m
                putMVar mv (s + 1)
        putMVar ms2 "ok"
```

- MVar ca semafor binar

```
newLock = newMVar ()      -- MVar care contine ()
aquireLock  m = takeMVar m
releaseLock  m = putMVar m ()
```

```
main = do
        m <- newLock
        forkIO $  forever (act1 m)
        forkIO $  forever (act2 m)
        getLine
```

forever repeta o actiune monadica de un numar infinit de ori

- MVar ca semafor binar

```
newLock = newMVar ()      -- MVar care contine ()
aquireLock  m = takeMVar m
releaseLock  m = putMVar m ()
```

```
act1 m = do
        aquireLock m
        print "I have the lock"
        releaseLock m
```

```
act2 m = do
        aquireLock m
        print "Now I am have the  lock"
        releaseLock m
```

```
main = do
        m <- newLock
        forkIO $  act1 m
        forkIO $  act2 m
        getLine
```

- MVar ca semafor binar

```
newLock = newMVar ()      -- MVar care contine ()
aquireLock  m = takeMVar m
releaseLock  m = putMVar m ()
```

```
main = do
        m <- newLock
        forkIO $  forever (act1 m)
        forkIO $  forever (act2 m)
        getLine
```

observam ca scrierea este sincronizata de m

```
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
"I have the lock"
"Now I have the lock"
```

➢ Sincronizarea accesului la stdout

```
(115.38 secs, 11,248,521,824 bytes)
*Main> main
FiFFFbiii bbb1  0233 005i  siii sss1  4147
711
2178309
24157817

""
(114.36 secs, 57,352 bytes)
```

Atentie!
Accesul la stdout nu este thread-safe, deci trebuie sincronizat

```
act n =  do
        let x = (fib n)
        putStrLn ("Fib " ++ (show n) ++ " is " ++ (show x))
main = do
      forkIO $ act 10
      forkIO $ act 20
      forkIO $ act 30
      forkIO $ act 35
      getLine
```

```
 -- threadSafeWrite
tsWrite stdo mes = do
            aquireLock stdo
            putStrLn mes
            releaseLock stdo

act n  stdo   =  do
        let x = (fib n)
        tsWrite stdo ("Fib " ++ (show n) ++
                          " is " ++ (show x))
main = do
        stdo <- newLock
        forkIO $ act 10 stdo
        forkIO $ act 20  stdo
        forkIO $ act 30   stdo
        forkIO $ act 35   stdo
        getLine
```

- Modelul Producator-Consumator
  MVar ca monitor

```
import Control.Concurrent
import Control.Monad


main = do
        m <- newEmptyMVar  --buffer
        forkIO (producer m )
        consumer  m 5 -- consuma 5 produse
        putStrLn "main thread ends"
```

- Modelul Producator-Consumator
  MVar ca monitor

```
import Control.Concurrent
import Control.Monad

main = do
        m <- newEmptyMVar  --buffer
        forkIO (producer m )
        consumer  m 5 -- consuma 5 produse
        putStrLn "main thread ends"
```

Thread-ul principal este consumator

```
producer :: MVar String-> IO ()
producer m = forever $ do
                                mes <- getLine
                                putMVar m mes
```

```
consumer :: MVar String -> Int  ->IO()
consumer m  n    =  if (n == 0)
                        then return ()
                        else
                              do
                                mes <- takeMVar m
                                putStrLn  (">"++ mes)
                                consumer m (n-1)
```

- Modelul Producator-Consumator
  MVar ca monitor

```
import Control.Concurrent
import Control.Monad
main = do
        m <- newEmptyMVar  --buffer
        l <- newLock
        forkIO (producer m )
        forkIO (consumer  m 10 l)
        releaseLock l
       putStrLn "main thread ends"
```

```
producer :: MVar String-> IO ()
producer m = forever $ do
                                mes <- getLine
                                putMVar m mes
```

```
consumer :: MVar String -> Int -> MVar () ->IO()
consumer m  n  l =  if (n == 0)
                        then aquireLock l
                        else
                              do
                                mes <- takeMVar m
                                putStrLn  (">"++ mes)
                                consumer m (n-1)  l
```

Thread-ul consumator se executa pe un thread nou. In acest caz, in thread-ul principal trebuie sa ne asiguram ca thread-ul consumator si-a terminat executia (folosind "lacatul" l).

- Modelul Producator-Consumator
  MVar ca monitor

```
*Main> main
m1
>m1
m2
>m2
m3
>m3
m4
>m4
m5
>m5
main thread ends
```

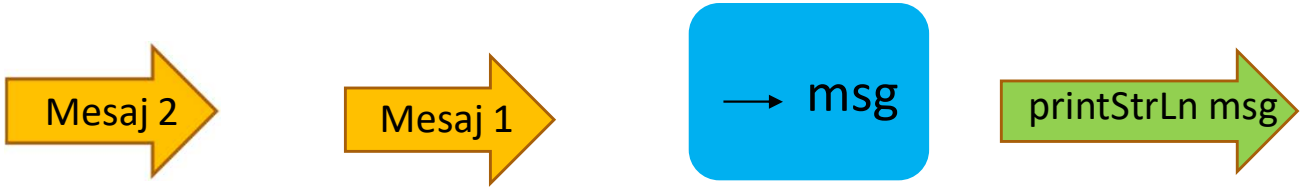
```
import Control.Concurrent
import Control.Monad

main = do
        m <- newEmptyMVar  --buffer
        l <- newLock
        forkIO (producer m )
        forkIO  (consumer  m 5 l)  -- consuma 5 produse
        releaseLock l
        putStrLn "main thread ends"
```

- Modelul Producator-Consumator

  Exemplu: serviciu de logare

  (modelarea unui canal de comunicare simplu folosind MVar)

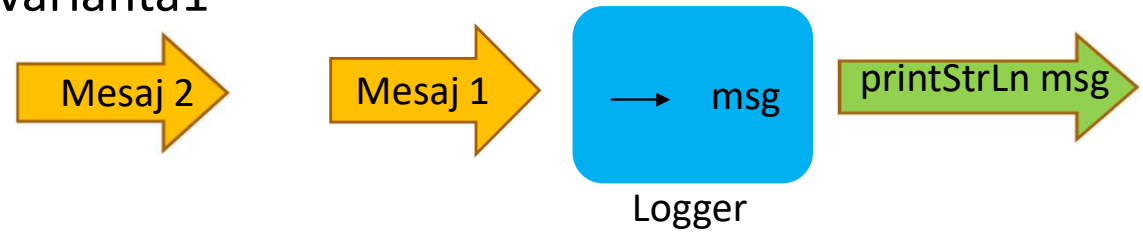
  http://chimera.labs.oreilly.com/books/1230000000929/ch07.html#sec_conc-logger



Cerinte:

- serviciul de logare prelucreaza mesajele intr-un thread separat
- mesajele trebuie prelucrate in ordinea in care sunt logate
- cand programul se termina toate mesajele logate trebuie sa fie prelucrate

Exemplu: serviciu de logare – varianta1

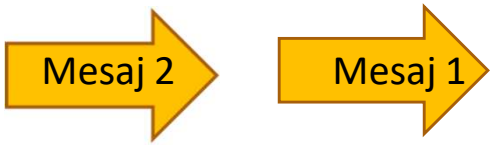



```
data Logger = Logger MVar  String
```

```
initLogger  ::  IO Logger
initLogger = do
                m <- newEmptyMVar

                let log = Logger m        -- log =

                forkIO (logger log)       -- creeaza

                return log

logger  :: Logger -> IO()  -- prelucreaza mesajele din Logger
```

```
logger :: Logger -> IO ()
logger (Logger m) = loop
                    where
                        loop = do
                                msg<- takeMVar m
                                putStrLn msg
                                loop
```

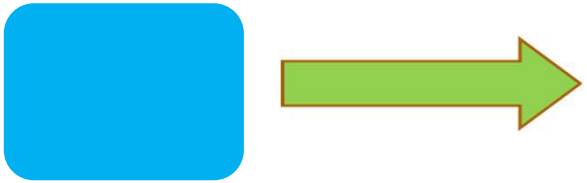
Exemplu: serviciu de logare- varianta1




```
logMessage :: Logger -> String -> IO ()
logMessage (Logger m) s = putMVar m s

logMessThread:: Logger -> IO()
logMessThread  l = do
                    msg <- getLine
                    if (msg == "bye")
                    then return()
                    else do
                          logMessage log msg
                          logMessTh log
main = do
          log <- initLogger
          logMessThread log
```

Creaza thread-ul logger

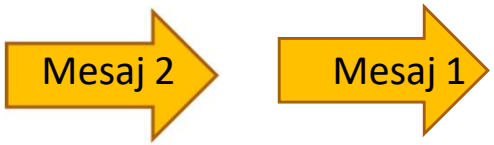
Thread-ul principal trimite messajele

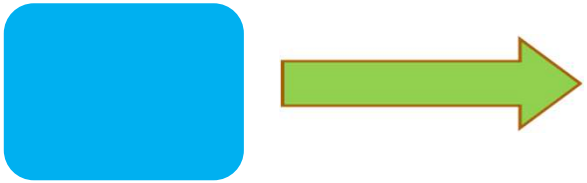


```
data Logger = Logger MVar  String

initLogger :: IO Logger
initLogger = do
                m <- newEmptyMVar
                let log = Logger m
                forkIO (logger log)
                return log

logger :: Logger -> IO ()
logger (Logger m) = loop
                        where
                            loop = do
                                    msg<- takeMVar m
                                    putStrLn msg
```

Thread-ul logger le citeste si le scrie

Exemplu: serviciu de logare- varianta1



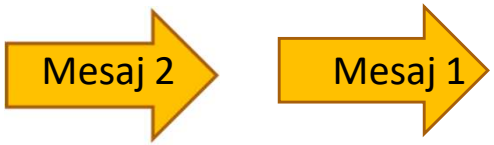


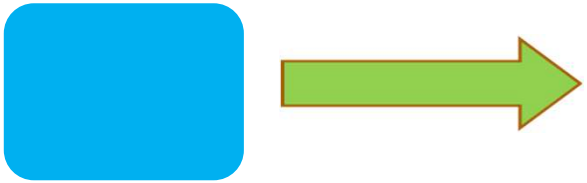
```
logMessage :: Logger -> String -> IO ()
logMessage (Logger m) s = putMVar m s

logMessThread:: Logger -> IO()
logMessThread log = do
                    msg <- getLine
                    if (msg == "bye")
                    then return()
                    else do
                          logMessage log msg
                          logMessTh log
main = do
          log <- initLogger
          logMessThread log
```

```
data Logger = Logger MVar  String

initLogger :: IO Logger
initLogger = do
                m <- newEmptyMVar
                let log = Logger m
                forkIO (logger log)
                return log
logger :: Logger -> IO ()
logger (Logger m) = loop
                    where
                        loop = do
                                msg<- takeMVar m
                                putStrLn msg
                                loop
```

Ce problema de sincronizare poate sa apara?

Exemplu: serviciu de logare- varianta1





```
logMessage :: Logger -> String -> IO ()
logMessage (Logger m) s = putMVar m s

logMessThread:: Logger -> IO()
logMessThread log = do
                    msg <- getLine
                    if (msg == "bye")
                    then return()
                    else do
                          logMessage log msg
                          logMessTh log
main = do
          log <- initLogger
          logMessThread log
```

```
data Logger = Logger MVar  String

initLogger :: IO Logger
initLogger = do
                m <- newEmptyMVar
                let log = Logger m
                forkIO (logger log)
                return log
logger :: Logger -> IO ()
logger (Logger m) = loop
                    where
                        loop = do
                                msg<- takeMVar m
                                putStrLn msg
                                loop
```

programul nu se asigura ca toate mesajele logate sunt prelucrate

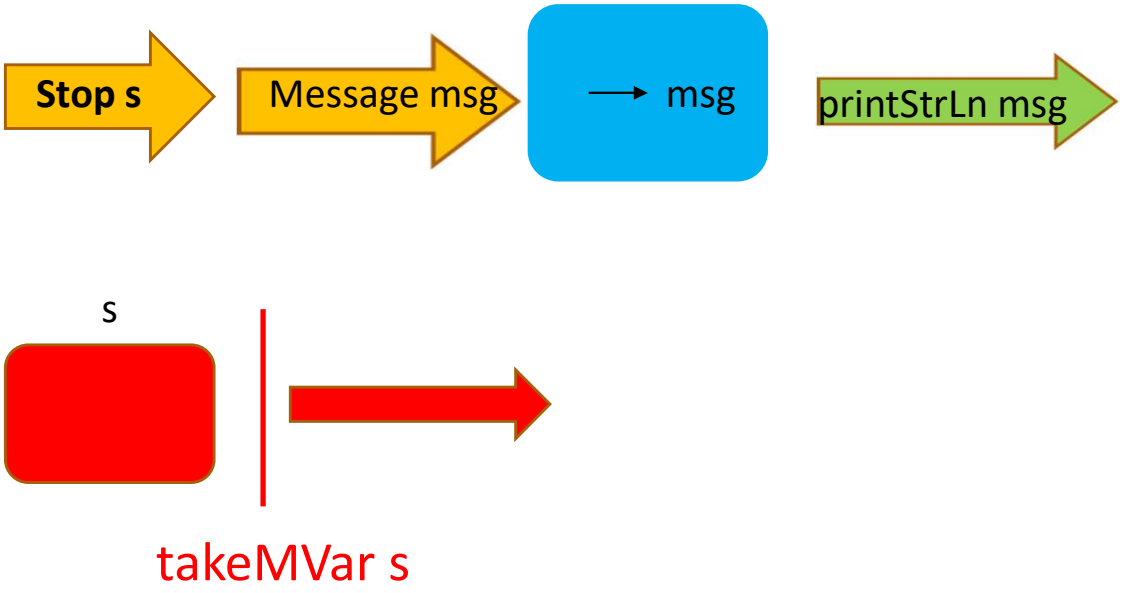
- Exemplu: serviciu de logare

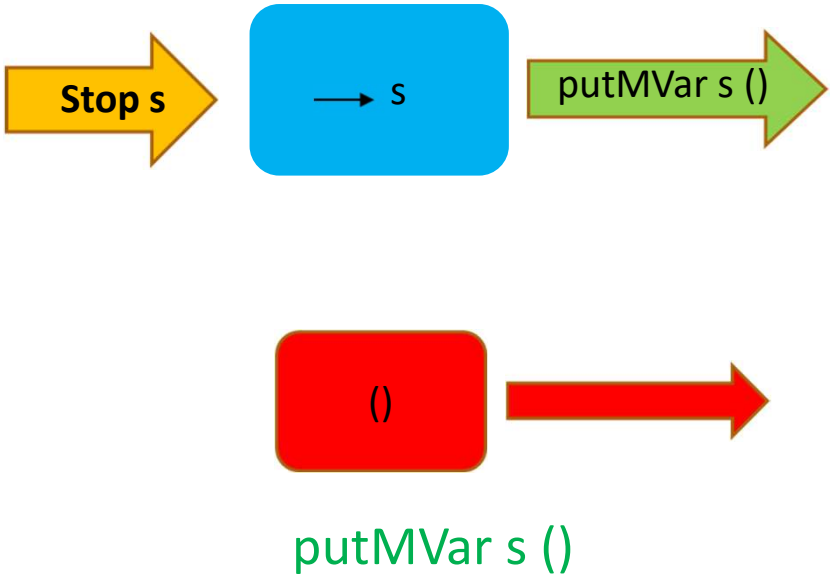
```
logMessage :: Logger -> String -> IO ()
logMessage (Logger m) s = putMVar m s

logMessThread:: Logger -> IO()
logMessThread log = do
                msg <- getLine
                if (msg == "bye")
                then logStop log
                else do
                      logMessage log msg
                      logMessTh log
```

```
-- la fel

initLogger :: IO Logger
initLogger = do
                m <- newEmptyMVar
                let log= Logger m
                forkIO (logger log)
                return log

main = do
            log <- initLogger
            logMessTh log
```

Exemplu: serviciu de logare

```
data Logger = Logger (MVar LogCommand)
data LogCommand = Message String | Stop  (MVar ())
```





- Exemplu: serviciu de logare

```
data Logger = Logger (MVar LogCommand)
data LogCommand = Message String | Stop  (MVar ())
```

```
logMessage :: Logger -> String -> IO ()
logMessage (Logger m) s = putMVar m s

logMessThread:: Logger -> IO()
logMessThread log = do
                msg <- getLine
                if (msg == "bye")
                then logStop log
                else do
                      logMessage log msg
                      logMessTh log
```

```
logStop :: Logger -> IO ()
logStop (Logger m) = do
                s <- newEmptyMVar
                putMVar m (Stop s)
                takeMVar s
```

```
logMessage :: Logger -> String -> IO ()
logMessage (Logger m) s = putMVar m (Message s)
```

Exemplu: serviciu de logare

```
logger :: Logger -> IO ()
logger (Logger m) = loop
            where  loop = do
                    cmd <- takeMVar m
                    case cmd of
                      Message msg -> do
                                putStrLn  ("mesaj: " ++ msg)
                                loop
                      Stop s -> do
                            putStrLn "logger: stop"
                            putMVar s ()
```

```
data Logger = Logger (MVar LogCommand)
data LogCommand = Message String | Stop  (MVar ())
```

Thread-ul logger va debloca s cand cand ajunge la Stop s

```
logStop :: Logger -> IO ()
logStop (Logger m) = do
            s <- newEmptyMVar
            putMVar m (Stop s)
            takeMVar s
```

logger.hs ©2012, Simon Marlow

```
*Main> main
mes:
mes1
mesm:e
saj: mes1
mes2
memseasj::
 mes2
mes3
mesm:e
saj: mes3
bye
```

```
stdo <- newMVar ()

tswrite stdo s = do
        takeMVar stdo
        putStrLn s
        putMVar stdo ()
```

```
*Main> main
mes:
mesajul 1
mesaj: mesajul 1
mes:
mesajul 2
mes:
mesaj: mesajul 2
mesajul 3
mes:
mesaj: mesajul 3
mesajul 4
mes:
mesaj: mesajul 4
bye
```